



HOMEWORK

Review on Chapter 1

- Classification of Ligands:
 - ✓ Based on σ -bonding:
lone pair donor, σ -bond electron pair donor, σ -bonding electron pair donor
 - ✓ Based on π -bonding:
 π donor, π acceptor
 - ✓ Most OM ligands are considered as π acceptors
- Notation for hapticity of ligands (η^n)
- Bonding involving lone pair donors, π -bond electron pair donors, and σ -bond electron pair
- Hard-soft acid/base
 - ✓ Hard acids like hard bases, soft acids like soft bases.
 - ✓ Most OM ligands are considered as soft bases.
- Other classifications of ligands
 - ✓ Based on charge: L and X types of ligands, e.g. NH_3 vs Cl^- .
 - ✓ Based on locations: Terminal and bridging ligands
- Valence electrons (VE) count
 - ✓ *Covalent model* : Both M and L are considered as neutral
 - ✓ *Ionic model* : Metal complexes are formed $\text{M}^{n+} + \text{L} + \text{X}^-$



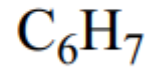
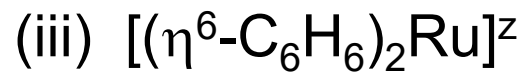
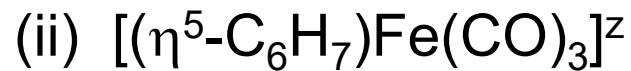
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- 18-electron rule and its limitation.
 - ***Not all OM complexes follow the rule.***
- Oxidation number
 - ***Oxidation number usually can not be higher than group member .***
- Coordination number and geometry
 - ***Each C.N. is associated with one or more geometries***
- Effect of complexation
 - ***Change of electron density of L***
 - ***Change of electron density distribution***
- Differences between metals
 - ***Trend in the relative energy of d orbitals***
- Trans effect and trans influence
 - ***Highest trans-effect ligands form unusually strong σ/π bonds***



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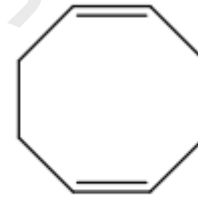
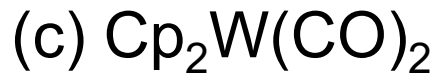
1. Based the 18-electron rule, determine the charge, z , of the following complexes.





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2. Propose a reasonable structure for each of the following empirical formula:

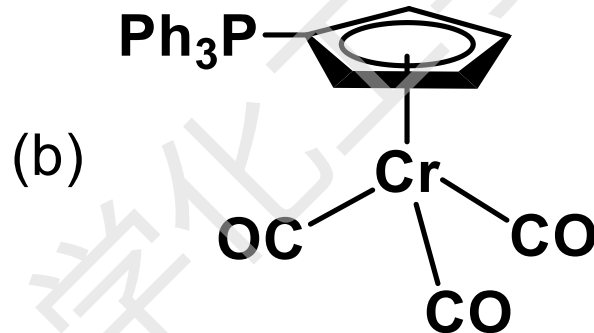
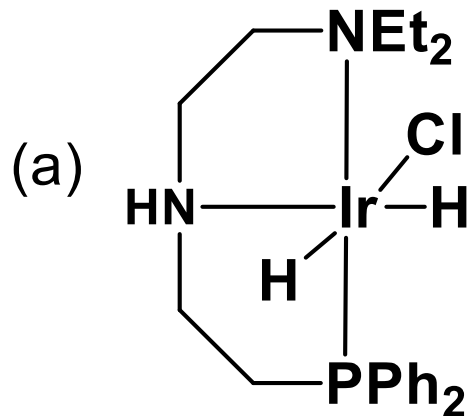


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3. Which of the following complexes follows/follow the 18e rule?



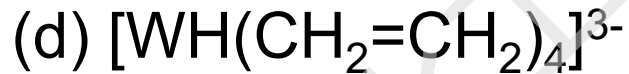
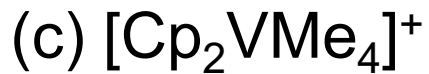
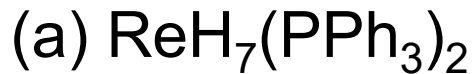
(A) both of them

(B) none of them



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4. Which of the following complexes is least likely to be isolated?

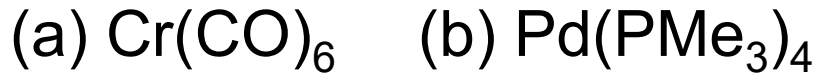


	3	4	5	6	7	8	9	10	11	12
21	22	23	24	25	26	27	28	29	30	
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	
39	40	41	42	43	44	45	46	47	48	
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	
57	72	73	74	75	76	77	78	79	80	
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	
89	104	105	106	107	108	109	110	111	112	
Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	



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5. On the basis of hard-soft acid-base concept, which of the following complexes would you expect to be least stable?





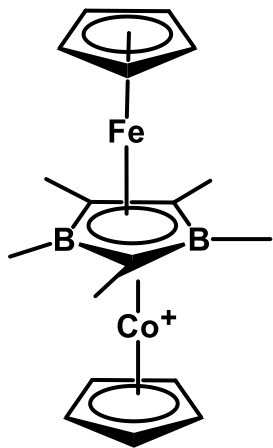
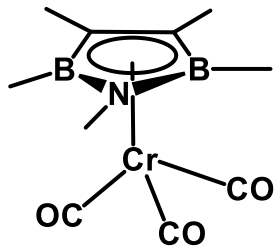
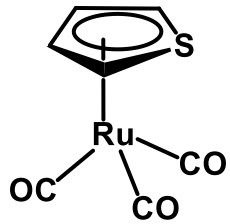
6. Which of the following complexes has the lowest $\nu(\text{C-O})$?





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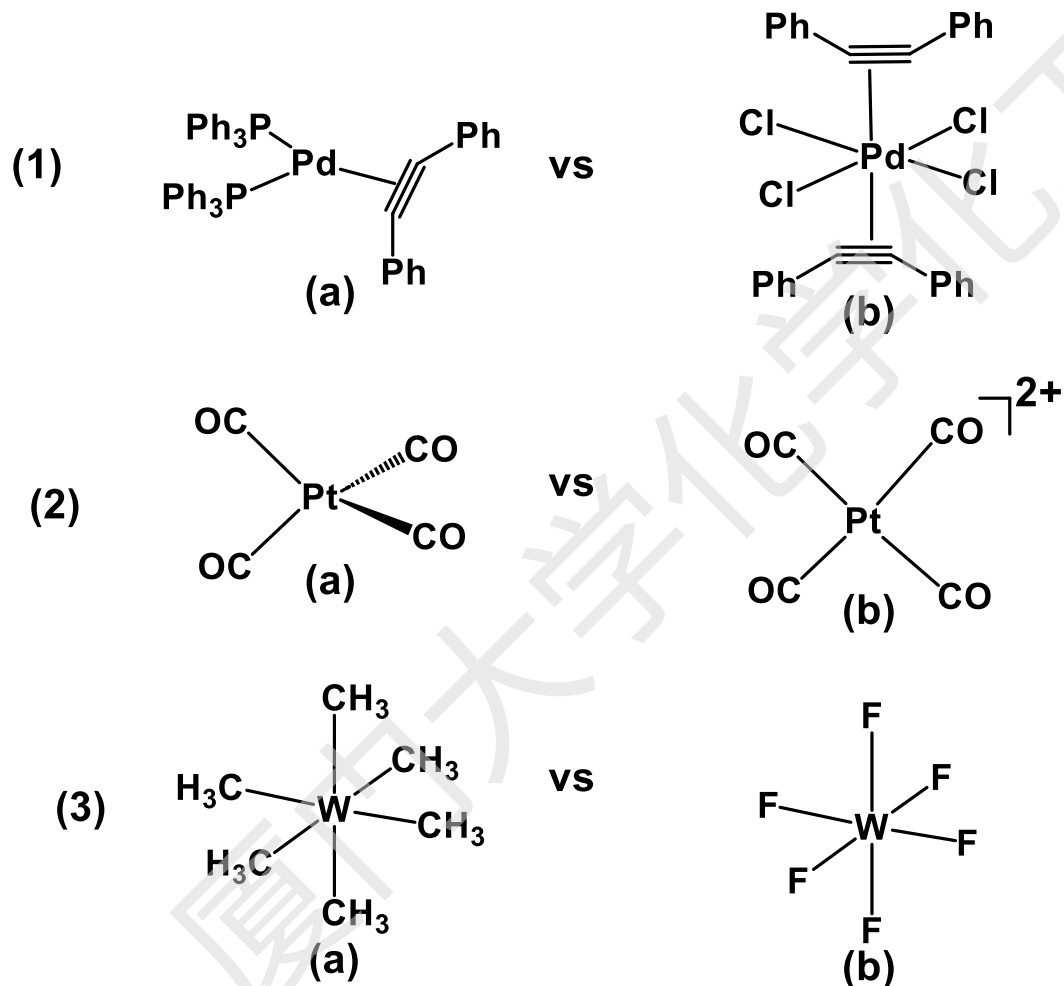
7. Do the following complexes follow the 18e rule?





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8. Which of the following complexes would you expect to be more stable?





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Bonus Challenge:

Read the short passage on "Differences between Metals" (1.9). Why is C-H activation (breaking a C-H bond) generally easier with second and third-row transition metals (like Rh, Ir, Pd, Pt) compared to first-row metals (like Ni, Co, Fe)?